# Will Kilgore

willkilgore@gmail.com | willkilgore.com

## Education

**McGill University** **Expected Dec 2027**

*Bachelor of Engineering in Mechanical Engineering - GPA: 3.84/4.0* *Montreal, QC*

## Work Experience

**Automated Tire, Inc.** **May 2026 – Aug 2026**

*Mechanical Engineering Intern – Autonomous tire-changing robotic platform* *Woburn, MA*

- Designed and fabricated a custom life-cycle testing jig, sourcing components and supplier quotes to evaluate tire-rotation mechanism durability under repeated loading
- Created technical drawings and assembly documentation in Onshape for 100+ component production assemblies, enabling technician-led builds without engineer oversight and reducing assembly time
- Designed, fabricated, and iterated process-specific fixtures and tooling, selecting manufacturing processes based on cost, lead time, and production volume to resolve assembly issues and reduce production time by 40%
- Simplified top-level assembly CAD by implementing a fastener-visibility toggle, reducing mate solving time from 3.5s to 1.2s (66% faster) across 4,000+ fastener instances, improving CAD responsiveness for design team
- Identified and documented design discrepancies during robotic system assembly, provided structured redline feedback, and processed engineering change orders that resolved recurring build issues across 2 design iterations
- Maintained three multi-material FDM printers, including calibration, hardware servicing, and failure diagnosis, reducing print failures and achieving same-day prototype turnaround for engineering team

**Parlee Composites** **June 2025 – Present**

*Manufacturing and Engineering Intern – Carbon fiber bicycle design and manufacturing* *Beverly, MA*

- Conducted experimental vibration analysis using accelerometer data and analytical models to evaluate bicycle frame compliance, informing design decisions for a new frame platform
- Developed a graphical fit-analysis system overlaying generated rider fit coordinates with 200 existing fits, reducing required production part count by 25% while maintaining rider coverage.
- Implemented an Excel-based fit analysis tool generating 150+ stem stack/reach coordinate combinations across varying stem lengths, stem angles, and frame sizes to reduce dealer consultation time
- Introduced new prepreg composite layups that reduced void content and post-processing time, increasing small-part production throughput by up to 30%
- Managed quality control for overseas frame production using inspection checklists and defect tracking, reducing recurring supplier issues through improved defect identification and feedback loops

## Projects

**McGill Rocket Team** **Sept 2023 – Present**

*Airframe Composites Project Co-Lead*

- Co-led composites manufacturing for the aerostructures subteam, coordinating layups, tooling, and assembly integration across a 10-member subteam to deliver flight-ready structures for Launch Canada
- Designed and fabricated molds for a 42” composite nose cone as part of the aerostructures subteam, enabling the transition to a fully student-built 8” rocket platform
- Executed vacuum-assisted resin infusion (VARI) and prepreg layups to manufacture lightweight, high-strength structural body tubes for competition rockets

## Skills

**Technical Skills:** Rapid Prototyping (3D Printing), Design for Manufacturing & Assembly (DFMA), GD&T, FEA, Vibrational Analysis, Experimental Design, Composite Manufacturing (Prepreg, VARI), Cable Routing & Harnessing

**Software:** CAD (SolidWorks, Onshape), Lifecycle Management (Arena), FEA (Abaqus), Python, Excel, AI-assisted workflows